A-A-53807
18 January 1989

COMMERCIAL ITEM DESCRIPTION

REGULATOR, PRESSURE, MEDICAL GAS ADMINISTRATION APPARATUS

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-R-36557C, which is cancelled.

This commercial item description covers the requirements for a single stage pressure regulator, with a Thorpe tube flowmeter suitable for use with oxygen cylinders and therapy equipments.

Salient characteristics:

General requirements. The regulator shall conform to the applicable requirements of CGA E-7, Standard for Flowmeters, Pressure Reducing Regulators, Regulator/Flowmeter and Regulator/Flowgauge combination for the Administration of Medical Gases, and UL 404, Standard for Safety, High Pressure Gas Gauges.

Order of precedence. In the event of a conflict between the requirements of CGA E-7 and UL 404, and the requirements described herein, the requirements specified in this document shall take precedence.

Material. The body of the regulator shall be fabricated from forged or machined copper alloy, nickel plated or chromium plated over nickel plated. There shall be no internal plating of the regulator body. Alloys shall be tempered and shall have a minimum tensile strength of 50,000 pounds per square inch. All external brass or copper alloy surfaces, such as gland nut, standard threaded fitting for low pressure oxygen connection and captive tubing adapter shall be nickel plated or chromium plated over nickel plated.

AMSC N/A FSC 6680

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Construction, style and design. The regulator shall consist essentially of a body, a pressure reducing valve, an inlet pressure gage, a Thorpe tube flow indicating device with a flow control valve for manually controlling the rate of low pressure flow, a standard externally-threaded fitting for connection to low pressure oxygen devices and a captive tubing adapter. The general style and design of the complete unit shall be in accordance with Figure 1.

Regulator. Regulator shall be of the single stage type which shall maintain uniform discharge rate by means of a sensitive diaphragm. The outlet pressure of the regulator shall be preset at 49 ± 1 pounds per square inch gage (psig). A self-reseating pressure relief valve shall be incorporated to prevent accumulation of hazardous pressure within the regulator which shall open at 130 psig maximum and reseat at 85 psig minimum. The regulator shall be equipped with a commercial union nut and gland inlet connection which shall be capable of making connection with suitable oxygen cylinders having a 0.903-14 NGO right-hand thread on the outlet valve in accordance with connection No. 540 of ANSI B57.1. The regulator inlet shall have a suitable assembled filter, fabricated from bronze, to prevent foreign matter from entering the device.

The regulator shall exhibit no leakage as evidenced by formation of bubbles, when a film of soapy water (grease free) is applied to all junctions of the regulator. The internal pressure applied to the regulator shall be at least 2000 psig and the flowmeter valve shall be completely closed (no output).

Pressure gage. The regulator shall be equipped with a pressure gage suitable for pressure measurement in the range of 0-4000 psig. The gage shall be calibrated to indicate both pressure and content. The pressure shall be marked at 100 psig and numbered at every 500 psig. The content shall be indicated as 1/4, 1/2, 3/4 and full. The face cover of the gage shall be fabricated from a clear, durable plastic.

Flow indicating device. The flow indicating device shall be a Thorpe tube flowmeter suitable for accurate measurement of oxygen flow in the range of 1 to 15 liters per minute (LPM). The flowmeter shall be calibrated in the range of 1 to 5 LPM on the lower half of the tube and 5 to 15 LPM in the upper half of the tube. At pressure of 50 psig and temperature of 70° Fahrenheit, the measurement shall be accurate to ± 1/2 LPM in the 1-5 LPM range and ± 1 LPM in the 5-15 LPM range.

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A clear flowmeter shield shall be provided over the flowtube. Both the flowtube and the shield shall be fabricated from impact-resistant plastic. The flowmeter shall possess a pressure relief device or feature that will prevent the flowmeter (flowtube) from bursting in the event of overpressurization.

The flowmeter valve, when tested at 0 degree and 110 degrees Fahrenheit, shall meet the following performance criteria:

a. Joints and shaft seals shall not leak when subjected to a pressure equal to 1.5 times the rated pressure of the flowmeter.

b. A shut-off torque of no more than 8 inch-pound shall be required to achieve a bubble tight seal at the outlet of the valve when the inlet of the valve is pressurized to the rated pressure of the flowmeter.

c. The valve shall be capable of sustaining 5000 cycles, and after the test, shall remain in compliance with the shut-off torque requirements. A cycle is defined as turning the valve to the off position and applying the torque sufficient to seal (but not to exceed 8 inch-pound), then turning the valve on until the maximum indicated flow is obtained.

d. The valve shall be capable of sustaining a tightening torque of 16 inch-pound without failing.

A removable captive tubing adapter shall be attached to the flowmeter. The tubing adapter shall be attached to a standard threaded oxygen fitting for low pressure application, in accordance with CGA connection No. 1240, as specified in CGA pamphlet V5. This standard fitting shall make a tight, leakproof connection with the regulator.

Workmanship. The regulator shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One regulator, as specified, constitutes one unit.

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Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 as specified herein. The unit of product for test purposes shall be one regulator.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document:

NSN	ITEM IDENTIFICATION
6680-01-174-6276	REGULATOR, PRESSURE, MEDICAL GAS ADMINSITRATION APPARATUS

Ordering data (Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order).

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD - MB

Agent:

DLA-DM

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CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
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Project No. 6680-0213

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-53807	REGULATOR, PRESSURE, MEDICAL GAS ADMINISTRATION APPARATUS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426

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In addition

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NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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